Supplementary Material: The MOND Interpolating Function from the Petz Recovery Bound

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We show that the Petz recovery fidelity bound $F \geq e^{-\Sigma/2}$ (Paper 1), combined with the observed Baryonic Tully–Fisher Relation (BTFR), *derives* the deep-MOND asymptotic form $\Sigma_{\rm gal} \to 2\sqrt{x}$ ($x = g_{\rm bar}/a_0$) without additional assumptions. Extending this to all x via the minimal (zero-parameter) extrapolation $\Sigma_{\rm gal} = 2\sqrt{x}$ yields the interpolating function $\mu(x) = 1 - e^{-\sqrt{x}}$, which coincides with the McGaugh–Lelli–Schombert empirical fit. The transition-region shape ($0.1 \lesssim x \lesssim 10$) is a *genuine prediction*—not a circular restatement of BTFR. We carefully separate what is derived from observation, what is ansatz (the minimal extrapolation), and what constitutes a falsifiable prediction, and identify the open problem of a microscopic derivation of $\Sigma_{\rm gal}$ from first principles.

I. THE PETZ RECOVERY BOUND

We recall the central result of Paper 1 [1]. For any completely positive trace-preserving (CPTP) channel $\mathcal{E}$ with entropy production

$$\Sigma \;=\; D\big(\rho \,\big\|\, \mathcal{R}_{\mathcal{E}}[\mathcal{E}(\rho)]\big) \geq 0\,, \tag{1}$$

the Petz recovery fidelity satisfies the Junge–Renner–Sutter–Wilde–Winter (JRSWW) bound [2]

$$F(\rho,\, \mathcal{R}_{\mathcal{E}}[\mathcal{E}(\rho)]) \;\geq\; e^{-\Sigma/2}\,. \tag{2}$$

Defining the temporal asymmetry parameter as the Petz recovery failure,

$$\tau \;=\; 1 - F\,, \tag{3}$$

the bound (2) yields an upper envelope

$$\tau_{\rm bound} \;=\; 1 - e^{-\Sigma/2}\,. \tag{4}$$

This is a universal, model-independent inequality: any physical process with entropy production Σ has its retrodictability failure bounded by Eq. (4). In Paper 1 we showed that this single equation, evaluated in different limits, reproduces the Crooks fluctuation theorem, Landauer's principle, and the gravitational Tolman–Oppenheimer–Volkoff redshift.

II. GALACTIC IDENTIFICATION

At galactic scales, the gravitational field produces an entropy associated with the deviation of the observed acceleration from the Newtonian prediction. We define the standard MOND ratio

$$x \;\equiv\; \frac{g_{\rm bar}}{a_0}\,, \tag{5}$$

where $g_{\rm bar} = GM_{\rm bar}/r^2$ is the Newtonian baryonic acceleration and $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is the MOND acceleration scale.

We identify the galactic gravitational entropy production as

$$\boxed{\Sigma_{\rm gal} \;=\; 2\sqrt{x} \;=\; 2\sqrt{\frac{g_{\rm bar}}{a_0}}} \tag{6}$$

Logical status of this identification. The deep-MOND asymptotic behavior $\Sigma_{\rm gal} \to 2\sqrt{x}$ as $x \to 0$ is *derived* from the observed BTFR combined with the Petz bound structure. The extension $\Sigma_{\rm gal} = 2\sqrt{x}$ to all x is the minimal (zero-parameter) extrapolation. The transition-region behavior of $\mu(x)$ is therefore a genuine prediction, not a circular restatement of BTFR. See Sec. VII for a precise separation of what is derived, what is an ansatz, and what constitutes a new prediction.

The physical motivation is as follows:

- In the *deep-MOND* regime ($x \to 0$): the gravitational field deviates maximally from Newton. $\Sigma_{\rm gal} \to 0$, reflecting that the Petz map can almost fully reconstruct the Newtonian state—the "retrodiction cost" per unit acceleration is small when there is almost no baryonic signal.

- In the *Newtonian* regime ($x \to \infty$): $\Sigma_{\rm gal} \to \infty$, so $\tau \to 1$ (saturation), meaning that the Newtonian dynamics is fully "irreversible" at the information-theoretic level—no MOND correction is needed.

The $\sqrt{x}$ form is the unique power law x^α that simultaneously gives $\mu \to 1$ as $x \to \infty$ (automatic for any $\alpha > 0$) and $\mu \to \sqrt{x}$ as $x \to 0$ (requires $\alpha = 1/2$; see Sec. IV).

III. DERIVATION OF THE INTERPOLATING FUNCTION

The MOND interpolating function $\mu(x)$ relates the observed gravitational acceleration $g_{\rm obs}$ to the baryonic

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Newtonian prediction:

$$g_{\rm obs} \;=\; \frac{g_{\rm bar}}{\mu(x)}\,. \tag{7}$$

In the τ framework, the fractional deviation of gravity from Newton *is* the Petz recovery failure: the inability to reconstruct the Newtonian gravitational state from the observed state quantifies precisely how much "extra" gravity is present. We therefore identify

$$\mu(x) \;=\; \tau_{\rm bound} \;=\; 1 - e^{-\Sigma_{\rm gal}/2}\,. \tag{8}$$

Substituting the galactic identification (6):

$$\boxed{\mu(x) \;=\; 1 - e^{-\sqrt{x}}} \tag{9}$$

This is the main result. The MOND interpolating function arises as the *saturated Petz recovery bound*—the boundary case where the gravitational channel is maximally efficient at converting entropy production into retrodiction failure. Note that:

1. No Boltzmann weights or thermal partition functions appear; this is a purely information-theoretic derivation.
2. The only inputs are the JRSWW bound (2) from quantum information theory and the identification (6) of the galactic entropy production.
3. The Crooks fluctuation theorem is *not* invoked. The derivation lives entirely within the Petz recovery framework.

IV. ASYMPTOTIC VERIFICATION

Newtonian limit ($x \to \infty$).—As $x \to \infty$, $e^{-\sqrt{x}} \to 0$, hence

$$\mu(x) \;\to\; 1\,, \tag{10}$$

recovering $g_{\rm obs} = g_{\rm bar}$: standard Newtonian gravity. ✓

Deep-MOND limit ($x \to 0$).—Expanding to leading order,

$$\mu(x) = 1 - e^{-\sqrt{x}} = \sqrt{x} - \frac{x}{2} + \frac{x^{3/2}}{6} - \cdots \;\approx\; \sqrt{x}\,. \tag{11}$$

Substituting into Eq. (7):

$$g_{\rm obs} \;=\; \frac{g_{\rm bar}}{\sqrt{g_{\rm bar}/a_0}} \;=\; \sqrt{g_{\rm bar}\,a_0}\,. \tag{12}$$

For a circular orbit, $g_{\rm obs} = V^2/r$ and $g_{\rm bar} = GM/r^2$, giving

$$V^4 \;=\; GM\,a_0\,, \tag{13}$$

the Baryonic Tully–Fisher Relation (BTFR) with zero intrinsic scatter. ✓

Transition region ($x \sim 1$).—At $x = 1$: $\mu(1) = 1 - e^{-1} \approx 0.632$. The transition from deep-MOND to Newtonian occurs smoothly over approximately one decade in x, centered near $x \sim 1$.

TABLE I. Comparison of MOND interpolating functions. $x = g_{\rm bar}/a_0$.

Name	$\mu(x)$	$\mu(1)$	$x \to 0$
Petz (this work)	$1 - e^{-\sqrt{x}}$	0.632	$\sqrt{x}$
Simple [5]	$\dfrac{x}{1+x}$	0.500	x
Standard [6]	$\dfrac{x}{\sqrt{1+x^2}}$	0.707	x
McGaugh RAR [4]	$1 - e^{-\sqrt{x}}$	0.632	$\sqrt{x}$

V. COMPARISON WITH OTHER INTERPOLATING FUNCTIONS

Table I compares the Petz-derived interpolating function with the standard choices in the MOND literature.

Several observations:

1. The Petz function (9) is *identical in functional form* to the empirical fit obtained by McGaugh, Lelli, and Schombert [4] from the SPARC Radial Acceleration Relation (RAR) data. This is a non-trivial consistency check: the information-theoretic derivation independently produces the same function that best fits 2693 data points from 153 galaxies.
2. The "simple" and "standard" interpolating functions have $\mu \sim x$ as $x \to 0$, which gives $g_{\rm obs} \sim g_{\rm bar}/x = a_0$—a constant acceleration floor. By contrast, the Petz function has $\mu \sim \sqrt{x}$, giving the physically correct $g_{\rm obs} = \sqrt{g_{\rm bar} a_0}$.
3. The transition sharpness differs: the Petz/McGaugh function transitions more gradually than the "standard" function ($\mu(1) = 0.632$ vs. 0.707) but more sharply than the "simple" function (0.632 vs. 0.500).
4. Since the Petz function matches the McGaugh RAR fit exactly, no additional fitting is needed against SPARC data. The agreement with observations is inherited directly.

VI. UNIQUENESS OF THE PETZ INTERPOLATING FUNCTION

We now prove that the interpolating function $\mu(x) = 1 - e^{-\sqrt{x}}$ is singled out by a combination of asymptotic, regularity, and minimality conditions. The results are organized in three stages of increasing strength.

A. Power-law uniqueness

Theorem 1 (Power-law uniqueness)**.** *Let $f\colon [0,\infty) \to [0,\infty)$ be a smooth function with $f(0) = 0$, and define $\mu(x) = 1 - e^{-f(x)/2}$. Suppose:*

(i) $\mu(x) \to \sqrt{x}$ *as* $x \to 0^+$ (deep-MOND / BTFR)*;*

(ii) $\mu(x) \to 1$ *as* $x \to \infty$ (Newtonian limit)*;*

(iii) $f(x) = c\,x^\alpha$ *for constants* $c > 0$, $\alpha > 0$ (power-law form)*.*

Then $\alpha = 1/2$ *and* $c = 2$*. That is,* $f(x) = 2\sqrt{x}$ *and* $\mu(x) = 1 - e^{-\sqrt{x}}$ *are the* unique *power-law solutions.*

Proof. **Step 1 (deep-MOND constraint).** For small $x > 0$, condition (iii) gives $f(x)/2 = (c/2)\,x^\alpha \to 0$. The Taylor expansion $1 - e^{-u} = u - u^2/2 + \cdots$ yields

$$\mu(x) \;=\; \frac{c}{2}\,x^\alpha - \frac{c^2}{8}\,x^{2\alpha} + O(x^{3\alpha})\,. \tag{14}$$

Condition (i) requires $\mu(x)/\sqrt{x} \to 1$ as $x \to 0^+$. Comparing leading powers: $x^\alpha \sim x^{1/2}$, so

$$\alpha = \frac{1}{2}\,. \tag{15}$$

Matching the coefficient: $(c/2)\,x^{1/2} \to \sqrt{x}$ requires $c/2 = 1$, hence

$$c = 2\,. \tag{16}$$

Step 2 (Newtonian limit check). With $f(x) = 2\sqrt{x}$: as $x \to \infty$, $e^{-\sqrt{x}} \to 0$, so $\mu(x) \to 1$. Condition (ii) is satisfied. ✓

Step 3 (uniqueness). Within the power-law family $f(x) = c\,x^\alpha$, the two constraints (i) and (ii) fix both parameters uniquely. No other member of this family satisfies both conditions. □

Remark. *If one relaxes to* $f(x) = c\,x^\alpha + d\,x^\beta$ *with* $0 < \alpha < \beta$*, condition (i) still forces* $\alpha = 1/2$ *and* $c = 2$ *(the subleading term* $d\,x^\beta$ *is irrelevant as* $x \to 0^+$*). Thus the deep-MOND constraint pins the leading monomial uniquely; subleading corrections are unconstrained by asymptotics alone.*

B. Strict concavity of the Petz interpolating function

The monotonicity and concavity of $\mu(x)$ are physically meaningful: monotonicity says that stronger baryonic acceleration produces dynamics closer to Newton; concavity says that the marginal return of increasing $g_{\rm bar}$ diminishes—a kind of "information-theoretic saturation."

Theorem 2 (Strict concavity)**.** *The function* $\mu(x) = 1 - e^{-\sqrt{x}}$ *satisfies:*

(a) $\mu'(x) > 0$ *for all* $x > 0$ (strictly increasing)*;*

(b) $\mu''(x) < 0$ *for all* $x > 0$ (strictly concave)*.*

Proof. We write $\mu(x) = 1 - e^{-x^{1/2}}$ and compute all derivatives explicitly.

First derivative. Let $u(x) = x^{1/2}$, so $u'(x) = \frac{1}{2}\,x^{-1/2}$. By the chain rule,

$$\mu'(x) \;=\; e^{-u}\,u' \;=\; \frac{e^{-\sqrt{x}}}{2\sqrt{x}}\,. \tag{17}$$

Since $e^{-\sqrt{x}} > 0$ and $\sqrt{x} > 0$ for all $x > 0$, we have $\mu'(x) > 0$. ✓

Second derivative. We differentiate $\mu'(x) = \frac{1}{2}\,x^{-1/2}\,e^{-x^{1/2}}$ using the product rule. Write

$$\mu'(x) = p(x)\,q(x)\,, \quad p(x) = \tfrac{1}{2}\,x^{-1/2}, \quad q(x) = e^{-x^{1/2}}\,. \tag{18}$$

Their individual derivatives are

$$p'(x) = -\frac{1}{4}\,x^{-3/2}\,, \tag{19}$$

$$q'(x) = e^{-x^{1/2}} \cdot \left(-\tfrac{1}{2}\,x^{-1/2}\right) = -\frac{e^{-\sqrt{x}}}{2\sqrt{x}}\,. \tag{20}$$

By the product rule, $\mu'' = p'\,q + p\,q'$:

$$\begin{aligned}\mu''(x) &= \left(-\frac{1}{4x^{3/2}}\right)e^{-\sqrt{x}} + \frac{1}{2\sqrt{x}} \cdot \left(-\frac{e^{-\sqrt{x}}}{2\sqrt{x}}\right) \\ &= -\frac{e^{-\sqrt{x}}}{4\,x^{3/2}} - \frac{e^{-\sqrt{x}}}{4\,x}\,.\end{aligned} \tag{21}$$

Factoring out $-e^{-\sqrt{x}}/4$:

$$\boxed{\mu''(x) \;=\; -\frac{e^{-\sqrt{x}}}{4}\left(\frac{1}{x^{3/2}} + \frac{1}{x}\right)} \tag{22}$$

For $x > 0$: $e^{-\sqrt{x}} > 0$, $x^{-3/2} > 0$, $x^{-1} > 0$, so the factor in parentheses is strictly positive, and the overall sign is negative. Therefore $\mu''(x) < 0$ for all $x > 0$. □

Remark. *The two terms in* (22) *have distinct origins. The* $x^{-3/2}$ *term comes from the curvature of the square-root mapping* $x \mapsto \sqrt{x}$ *(i.e. from* p'*), while the* x^{-1} *term comes from the exponential saturation of the Petz bound (i.e. from* q'*). Both contribute with the same sign, so concavity is* robust*: it does not rely on a cancellation.*

C. Characterization theorem

The following theorem combines the results above into a single characterization.

Theorem 3 (Characterization)**.** *Let* $f\colon [0,\infty) \to [0,\infty)$ *be smooth and monotonically increasing with* $f(0) = 0$*, and define* $\mu_f(x) = 1 - e^{-f(x)/2}$*. Suppose:*

(i) $\mu_f(x)/\sqrt{x} \to 1$ *as* $x \to 0^+$ (BTFR)*;*

(ii) $\mu_f(x) \to 1$ *as* $x \to \infty$ (Newtonian limit).

Then:

1. ***Asymptotic form.*** $f(x) = 2\sqrt{x} + h(x)$, *where* $h(x)/\sqrt{x} \to 0$ *as* $x \to 0^+$.

2. ***Monomial uniqueness.*** *If* f *is a monomial* ($f(x) = c\,x^\alpha$), *then* $f(x) = 2\sqrt{x}$.

3. ***Concavity of the minimal solution.*** *The minimal (zero-parameter) choice* $h = 0$, *giving* $\mu(x) = 1 - e^{-\sqrt{x}}$, *is strictly concave on* $(0, \infty)$.

4. ***Zero-parameter uniqueness.*** *Among all functions of the form* $\mu_f(x) = 1 - e^{-f(x)/2}$ *satisfying* (i) *and* (ii), *the choice* $f(x) = 2\sqrt{x}$ *is the unique member with no free parameters beyond those fixed by* (i) *and* (ii).

Proof. **Part 1.** Condition (i) with $f(x) \to 0$ gives $\mu_f(x) \approx f(x)/2 + O(f^2)$. Hence $f(x)/2 \to \sqrt{x}$ as $x \to 0^+$, i.e. $f(x) = 2\sqrt{x} + h(x)$ with $h(x)/\sqrt{x} \to 0$.

Part 2. If $f(x) = c\,x^\alpha$, Part 1 forces the leading behavior $c\,x^\alpha \sim 2\,x^{1/2}$, so $\alpha = 1/2$ and $c = 2$. This is Theorem 1.

Part 3. This is Theorem 2.

Part 4. Any f satisfying Part 1 takes the form $f(x) = 2\sqrt{x} + h(x)$ with $h(x)/\sqrt{x} \to 0$. The correction h is a free function (subject to regularity). Setting $h = 0$ is the unique choice that introduces no additional functional freedom. Every other choice requires specifying the form of h (e.g. $h(x) = d\,x^\beta$ with $\beta > 1/2$), which introduces at least one free parameter. □

Corollary 4. *If one imposes Occam's razor—the interpolating function* $\mu(x) = 1 - e^{-f(x)/2}$ *should involve no free parameters beyond those fixed by the BTFR and Newtonian asymptotics—then* $\mu(x) = 1 - e^{-\sqrt{x}}$ *is the unique solution. This zero-parameter function is automatically strictly monotone and strictly concave.*

VII. WHAT REMAINS CONJECTURAL

It is important to clearly separate what is derived from observation, what is an extrapolation, and what constitutes a genuinely new prediction. We organize the logical status of each element below.

A. Derived from BTFR (not conjectural)

The deep-MOND asymptotic form $\Sigma_{\rm gal} \to 2\sqrt{x}$ as $x \to 0$ is [DERIVED], not [ASSUMED]. The argument is as follows. Within the Petz bound framework, $\mu(x) = 1 - e^{-\Sigma_{\rm gal}/2}$. In the deep-MOND limit ($x \to 0$), $\Sigma_{\rm gal} \to 0$ and $\mu \approx \Sigma_{\rm gal}/2$. The BTFR ($V^4 = GMa_0$) requires $\mu \to \sqrt{x}$ as $x \to 0$ (see Eq. (13)). Combining: $\Sigma_{\rm gal}/2 \to \sqrt{x}$, i.e. $\Sigma_{\rm gal} \to 2\sqrt{x}$.

This step uses only: (a) the JRSWW bound structure, and (b) the empirically established BTFR. It is a *derivation*, not a fit.

B. Ansatz: minimal zero-parameter extrapolation

The extension

$$\Sigma_{\rm gal} = 2\sqrt{x} \quad \text{for } \textit{all } x \tag{6}$$

is the simplest (zero-parameter) extrapolation of the derived deep-MOND asymptotic form to the full range of x. This is an [ASSUMED] ansatz, not a derivation. More complex functional forms—e.g. $\Sigma_{\rm gal} = 2\sqrt{x}\,h(x)$ with $h(x) \to 1$ as $x \to 0$—could in principle modify the transition region while preserving the BTFR. The choice $h(x) = 1$ (no additional parameters) is the minimal hypothesis.

Three supporting arguments for this minimal choice:

(i) Dimensional consistency. Σ must be dimensionless. The combination $g_{\rm bar}/a_0$ is dimensionless, and its square root is the simplest function thereof. The prefactor 2 is fixed by matching the JRSWW bound normalization (2).

(ii) Power-law uniqueness. Consider the general power-law ansatz $\Sigma_{\rm gal} = 2\,x^\alpha$. Then $\mu = 1 - e^{-x^\alpha}$. The deep-MOND limit $\mu \approx x^\alpha$ must equal $\sqrt{x}$ to reproduce the BTFR (13), forcing $\alpha = 1/2$. No other power law works. (Note: this fixes the *exponent* but does not exclude non-power-law corrections at finite x.)

(iii) Connection to the cosmological sector. If $\mu_0 = H_0/c$ (as suggested by Paper 3's cosmological analysis) and $a_0 = cH_0/(2\pi)$ (Milgrom's coincidence), then

$$\Sigma_{\rm gal} = 2\sqrt{\frac{g_{\rm bar}}{c^2\mu_0/(2\pi)}}\,, \tag{23}$$

linking the galactic entropy production to the same parameter μ_0 that governs the cosmological sector.

C. Genuinely new prediction

Given the minimal ansatz $\Sigma_{\rm gal} = 2\sqrt{x}$ for all x, the interpolating function $\mu(x) = 1 - e^{-\sqrt{x}}$ makes a *genuine prediction* for the transition-region behavior at $0.1 \lesssim x \lesssim 10$, where neither the deep-MOND nor the Newtonian asymptotic applies. This prediction is *not* a circular restatement of BTFR (which constrains only $x \to 0$) nor of Newtonian gravity (which constrains only $x \to \infty$). The transition shape—in particular, $\mu(1) = 1 - e^{-1} \approx 0.632$—is testable against SPARC RAR data and constitutes the falsifiable content of the ansatz.

The fact that this predicted shape coincides with the McGaugh et al. [4] empirical fit to 2693 data points from 153 galaxies is a nontrivial success of the minimal extrapolation, not a built-in feature.

D. Open problem: microscopic derivation

A microscopic derivation would require computing the quantum relative entropy $D(\rho_{\rm obs}\|\rho_{\rm Newton})$ for the gravitational field in the weak-field, quasi-static limit, and showing that $\Sigma_{\rm gal} = 2\sqrt{x}$ emerges without the minimal-extrapolation assumption. This would promote the ansatz to a theorem. We leave this to future work.

VIII. SUMMARY

The derivation chain is:

$$\underbrace{F \geq e^{-\Sigma/2}}_{\text{JRSWW bound}} \xrightarrow{\Sigma_{\rm gal}=2\sqrt{x}} \underbrace{\mu(x) = 1 - e^{-\sqrt{x}}}_{\text{MOND interpolation}} \tag{24}$$

The Petz recovery bound, combined with a single identification of the galactic entropy production, yields the MOND interpolating function that coincides with the best empirical fit to the Radial Acceleration Relation. The derivation uses no Boltzmann weights, no Crooks theorem, and no partition functions—only the information-theoretic structure of quantum channels.

ACKNOWLEDGMENTS

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